

VERKHNETULOMSKIY, Russian Federation

WMO: 221060

Lat: 68.6083N

Lon: 31.8039E

Elev: 206

StdP: 14.59

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.5	-22.4	-33.3	0.8	-27.9	-26.8	1.2	-22.3	12.6	24.0	10.9	22.8	0.1	160	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	76.5	62.8	72.2	60.6	68.4	58.6	64.7	73.6	62.0	69.9	59.7	67.2	4.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.8	80.3	69.9	58.4	73.4	66.7	56.0	67.2	63.5	29.8	74.3	27.9	70.1	26.3	67.0	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.3	7.8	-33.1	82.0	6.7	3.4	-37.9	84.4	-41.8	86.5	-45.6	88.4	-50.5	90.9		
(5)			-33.0	67.4	6.4	2.7	-37.6	69.3	-41.3	70.9	-44.9	72.4	-49.5	74.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.3	8.7	9.4	20.4	31.4	40.6	50.1	57.0	53.0	45.1	33.7	21.6	15.3		
	DBStd	19.24	15.32	14.86	10.51	7.70	7.40	7.50	7.18	6.31	6.36	7.98	11.42	13.37		
	HDD50	6950	1279	1137	917	558	307	89	18	37	175	505	852	1076		
	HDD65	11945	1744	1557	1382	1008	755	450	263	376	598	969	1302	1541		
	CDD50	500	0	0	0	0	17	92	235	130	26	0	0	0		
	CDD65	24	0	0	0	0	0	3	16	5	0	0	0	0		
	CDH74	229	0	0	0	0	4	30	159	36	0	0	0	0		
	CDH80	33	0	0	0	0	1	3	26	3	0	0	0	0		
(14) Wind	WSAvg	3.3	2.7	2.9	3.7	3.7	4.0	4.2	3.6	3.0	3.3	3.4	2.8	2.8		
(15) Precipitation	PrecAvg	21.1	1.3	1.1	1.1	1.2	1.6	2.4	2.9	2.5	2.0	2.1	1.4	1.5		
	PrecMax	26.0	2.5	1.9	2.4	3.1	2.4	4.9	5.3	5.7	4.0	4.7	2.7	2.6		
	PrecMin	16.1	0.6	0.4	0.3	0.4	0.6	0.4	1.0	0.8	0.5	0.6	0.4	0.6		
	PrecStd	2.8	0.4	0.4	0.4	0.6	0.6	1.2	1.1	1.2	0.9	1.0	0.5	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.5	37.9	41.9	54.7	70.7	78.7	83.6	78.6	65.2	52.7	42.8	38.7		
		MCWB	33.7	35.1	36.2	44.4	56.1	63.9	66.0	64.5	57.1	48.0	40.8	35.8		
	2%	DB	33.0	34.7	38.2	47.9	63.2	72.2	78.3	72.7	60.8	49.5	38.6	36.5		
		MCWB	31.4	32.3	34.0	40.5	51.4	59.0	63.7	62.1	54.7	45.8	36.8	34.1		
	5%	DB	30.2	32.3	35.5	44.6	58.3	68.2	74.4	67.9	57.5	46.8	36.6	33.9		
		MCWB	28.8	30.1	31.9	38.4	48.9	56.6	61.8	59.3	52.9	44.0	34.9	32.1		
	10%	DB	27.2	29.2	33.4	41.7	54.0	64.3	70.4	63.9	54.5	44.2	34.5	31.1		
		MCWB	26.1	27.5	30.6	36.6	46.3	54.6	60.3	57.7	50.8	42.0	33.1	29.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.1	35.5	36.8	45.0	58.3	66.5	68.4	66.0	58.2	48.6	41.2	36.0		
		MCDB	36.0	37.6	40.9	54.0	68.8	76.2	77.9	75.9	63.7	51.9	42.8	38.0		
	2%	WB	31.5	32.6	34.3	41.1	52.7	60.7	65.6	63.2	55.3	46.4	37.1	34.3		
		MCDB	32.8	34.5	37.4	46.8	61.7	69.5	75.1	70.5	59.6	49.2	38.4	36.3		
	5%	WB	28.9	30.1	32.5	39.0	49.5	57.5	63.2	60.4	53.1	44.1	35.0	32.2		
		MCDB	30.0	32.1	34.9	43.7	57.2	66.2	71.9	66.8	57.0	46.4	36.3	33.8		
	10%	WB	26.2	27.6	30.8	37.0	46.7	55.2	61.0	58.0	51.0	42.1	33.2	29.7		
		MCDB	27.2	29.2	33.3	41.1	53.6	63.8	69.2	63.4	54.4	44.1	34.4	31.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	14.0	15.1	13.8	13.1	14.4	15.0	13.6	11.4	7.6	9.4	11.3		
		MCDBR	10.7	9.1	12.0	17.0	22.9	21.7	23.5	21.6	16.4	8.6	6.7	8.4		
	5% WB	MCWBR	9.9	7.8	9.2	10.6	13.3	11.1	10.9	11.5	10.5	6.6	6.3	7.6		
		MCWBR	10.2	9.0	10.6	15.3	21.0	19.3	19.4	17.9	14.9	7.9	6.5	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.206	0.257	0.302	0.335	0.358	0.355	0.372	0.354	0.319	0.278	0.360	N/A		
	taud		1.913	2.049	2.100	2.155	2.336	2.431	2.412	2.437	2.503	2.244	2.669	N/A		
	Ebn at Noon		17	159	223	253	263	268	258	249	230	154	30	0		
	Edh at Noon		2	15	27	34	32	30	30	26	19	12	3	0		
(44) All-Sky Solar Radiation	RadAvg		8	137	574	1093	1390	1519	1395	961	526	158	22	0		
	RadStd		1	16	35	71	144	135	171	100	49	22	3	0		

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (21 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page